

Predictive Analytic using Multiple Linear Regression

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Predicting MEDV housing price

This dataset contains information collected by the US Census Service concerning housing in the area of Boston Massachusetts. The dataset has 506 cases. The data was originally published by Harrison, D. and Rubinfeld, D.L. `Hedonic prices and the demand for clean air', J. Environ. Economics & Management, vol.5, 81-102, 1978.

The file Boston.xls contains the following 14 attributes:

CRIM	per capita crime rate by town
ZN	proportion of residential land zoned for lots over 25,000 sq.ft.
INDUS	proportion of non-retail business acres per town.
CHAS	Charles River dummy variable (1 if tract bounds river; 0 otherwise)
NOX	nitric oxides concentration (parts per 10 million)
RM	average number of rooms per dwelling
AGE	proportion of owner-occupied units built prior to 1940
DIS	weighted distances to five Boston employment centres
RAD	index of accessibility to radial highways
TAX	full-value property-tax rate per $10,000
PTRATIO	pupil-teacher ratio by town
B	1000(Bk - 0.63)^2 where Bk is the proportion of blacks by town
LSTAT	% lower status of the population
MEDV	Median value of owner-occupied homes in $1000

Answer the following questions

a) Explore (EDA) the dataset, visualize the correlation matrix using Heatmap.

b) Identify two attributes having highest correlation with MEDV.

c) Construct a predictive model using linear regression to predict MEDV based on the two attributes.

- Use hold-out method with testing size = 20%
- Scale your training and testing data using StandardScaler.
- Write the regression equation.
- Report RMSE

d) [Extra-Points] Improve accuracy of model by decreasing error.

- Write the regression equation.
- Report RMSE